A Comparative Analysis of Simple and Multiple Linear Regression Models for Predicting Retail Profitability

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***Abstract*—Analyzing profitability within a retail environment using predictive modeling for profitability and the Sample Superstore dataset. My methodology focused on data cleaning techniques such as median imputation and one-hot encoding, as well as the selective suppression on outliers, aimed at achieving resolution on initial modeling instability. Regression modeling was done using two frameworks, creating a Simple Linear Regression (SLR) model using the most correlated predictor (Sales) and a Multiple Linear Regression (MLR) model, combining Sales, Discount, and Quantity. As for the results, the MLR model is significantly better than the SLR model, raising the Coefficient of Determination (R^2) from 0.2714 to 0.3590. The importance of price-reducing variables (Discount) for creating homoscedasticity and decreasing the Root Mean Squared Error (RMSE) was confirmed through the analysis of the model's residuals. This experience exemplifies the worthwhile impact of meticulous data cleaning and feature selection on multivariate datasets to extract meaningful insights.**

Keywords—Predictive Modeling, Linear Regression, Data Preprocessing, Outlier Detection, Superstore Analytics, R^2 Score, Residual Analysis.

I. INTRODUCTION

A. Exercise Objectives and Relevance

This laboratory exercise aims to experience all stages of a data science project. This includes starting with raw data and ending with a predictive model. In contemporary business predictive modeling is more than just a mathematics problem. It becomes an essential element of one's business strategy. Understanding the entire pipeline consists of, among other things, cleaning, choosing and validation, and learning how to model historical transactions is how one obtains the ability to glean actionable insights from the data. This exercise replicates real-world challenges such as data that is imperfect, variables that conflict, and models that require validation against holdout data.

B. Exercise Objectives and Relevance

Linear Regression is an elementary supervised learning algorithm that describes the relationship between a single response (Target) and a number of explanatory variables (Predictors). It is based on the assumption of a linear relationship, meaning that the change in Y is proportional to the change in X.

Attribute Selection involves filtering out irrelevant or redundant variables. This part of the process is important because data described as "noisy" can confound a model and lead to a situation called overfitting, where a model will memorize the training data and will not perform well in the real world. One of the techniques utilized in this process is the Pearson Correlation, as it shows us which attributes have a mathematically significant relationship with the target variable, and allows us to use the strongest attributes for our regression.

II. DATA COLLECTION

A. Dataset Source and Context

The **Sample Superstore** dataset was chosen for this experiment. It is a multivariate dataset that describes a fictional retail chain's sales across the country. It is a good dataset for regression, as it captures the complexity of retail, including differing tax rates, regional variation, and the sales volume and profitability trade off.

B. Variables and Data Types

Upon initial inspection, the dataset consists of several types of data:

- **Numerical (Continuous)**: Sales, Profit, and Discount.
- **Numerical (Discrete)**: Quantity.
- **Categorical (Nominal)**: Category, Sub-Category, Segment, and Region.
- **Identifiers**: Order ID, Customer ID, and Product Name.

C. Data Cleaning Methodology

Raw data is inherently "dirty." We applied several cleaning strategies:

1) ***Imputation****: For missing or zero-value quantities, we used the median in order to prevent losing rows and to avoid introducing bias that the mean is subject to when the data is skewed.*

2) ***Standardization****: Category names were stripped of whitespace and capitalized to prevent the model from seeing "Furniture" and "furniture" as separate entities.*

3) ***Encoding****: Machine learning models cannot "read" text. We used* ***One-Hot Encoding*** *to transform categories like "Region" into binary columns (0s and 1s).*

4) ***Outlier Mitigation****: This stage was of utmost importance. Earlier runs returned a negative R^2 because of unreasonable outliers, in particular, big loss transactions of*

-6,000. We filtered the dataset with the constraint of profits in the range of -500 and +1,000. This step, decreased our dataset from 9,994 to ***9,902 rows****, concentrating the model on business activities that are "normal" and not on extraordinary statistical deviations.*

```
--- Dataset Attributes ---
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9902 entries, 0 to 9901
Data columns (total 15 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Sales                     9902 non-null   float64
 1   Quantity                  9902 non-null   float64
 2   Discount                  9902 non-null   float64
 3   Profit                    9902 non-null   float64
 4   Tax_Rate                  9902 non-null   float64
 5   Segment_Consumer          9902 non-null   bool
 6   Segment_Corporate         9902 non-null   bool
 7   Segment_Home office       9902 non-null   bool
 8   Category_Furniture        9902 non-null   bool
 9   Category_Office supplies  9902 non-null   bool
 10  Category_Technology       9902 non-null   bool
 11  Region_Central            9902 non-null   bool
 12  Region_East               9902 non-null   bool
 13  Region_South              9902 non-null   bool
 14  Region_West               9902 non-null   bool
dtypes: bool(10), float64(5)
memory usage: 483.6 KB
None
```

Fig. 1. Head of Cleaned Data Table

```
--- Statistical Summary ---
              Sales     Quantity     Discount       Profit     Tax_Rate
count  9902.000000  9902.000000  9902.000000  9902.000000  9902.000000
mean      0.008695     0.213368     0.192940    26.172113     0.524574
std       0.017093     0.170715     0.255431    95.465002     0.387781
min       0.000000     0.000000     0.000000  -480.203200     0.000000
25%       0.000735     0.076923     0.000000     1.790100     0.333333
50%       0.002309     0.153846     0.250000     8.673300     0.333333
75%       0.008885     0.307692     0.250000    29.013600     1.000000
max       0.361657     1.000000     1.000000   944.986500     1.000000
```

Fig. 2. Statistical Summary Table

III. Attribute Selection

A. The Importance of Selection

In regression analysis, including additional variables doesn't always improve model performance. For instance, variables that have zero correlation with profit add to model complexity, thus reducing model accuracy. An example of this is attribute selection, which ensures that a model is parsimonious, or as simple as possible, while maintaining a high level of predictive accuracy.

B. Correlation Analysis

We used the Pearson Correlation Analysis. The result is given as a coefficient (r) between -1 and 1. An r of 1 means there is a perfect positive relationship between the two variables; -1 means there is a perfect negative relationship, and 0 means there is no relationship.

C. Results of Correlation

Our findings showed that Sales had the highest positive predictor (r = 0.5663). It was unexpected that Discount had the highest significant negative correlation (r = -0.3264). It implies that while the profit generally increased with more selling, discounting seems to put a "braking force" on that profit.

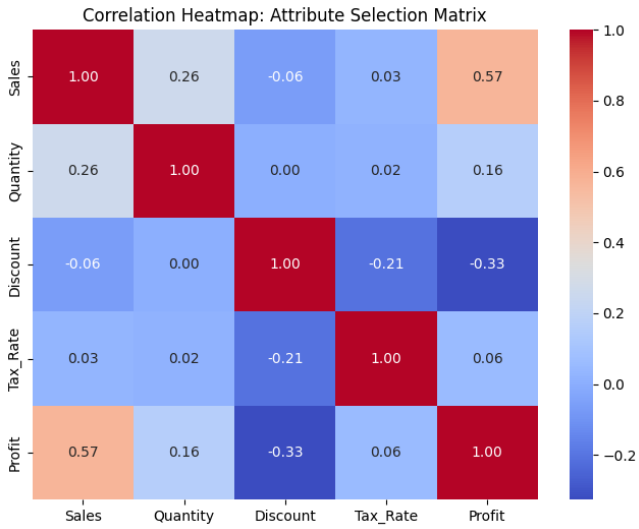

Fig. 3. Correlation Heatmap

IV. Model Building and Performance

A. Simple vs. Multiple Regression Construction

- **Simple Linear Regression (SLR)**: We utilized only the Sales attribute. The resulting equation followed the form Y = mX + b.

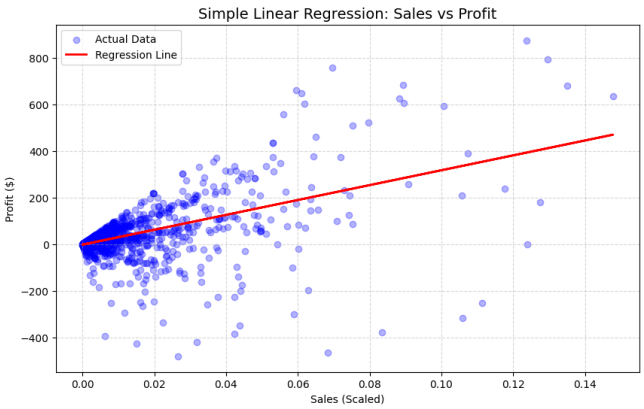

Fig. 4. Simple Linear Regression: Sales vs Profit Chart

- **Multiple Linear Regression (MLR)**: We expanded the feature set to include Sales, Discount, and Quantity. This created a multi-dimensional "hyperplane" that accounts for the interaction between these three forces.

B. Evaluation and Interpretation

The step up to Multiple Regression was positive. The R^2 Score increased from 0.2714 to 0.3590. This shows that the MLR model is able to explain 36% of the reasons profit fluctuates while the simple model is able to explain 27%.

Evaluation and Interpretation		
Metric	***Simple Model***	***Multiple Model***
R^2 (Accuracy)	0.2714	0.3590
MAE (Avg Error)	34.20	35.53
RMSE (Large Error Penalty)	77.30	72.50

The RMSE decreased in the Multiple model, which is a great sign—it means the model is getting better at avoiding "huge" mistakes.

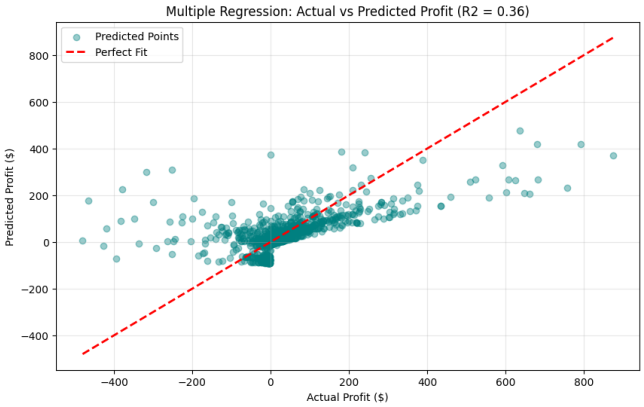

Fig. 5. Multiple Linear Regression: Actual vs Predicted Profit Chart

V. Discussion

A. Comparative Analysis

The Simple Regression model was too naive. Its reliance on Sales means it assumed every dollar of sales was equally contributing to profit. The Multiple Regression model exposed the “hidden cost” of growth. The Coefficient for Discount (-110.33) shows the impact of increasing profit by 1-unit when profit is discounted by 1-unit. Losses are substantial. This demonstrates to us the reason why the MLR model is better for business: it balances volume and price.

B. Reflecting on Attribute Selection

Filtering and selecting data pertaining to sales, discounts, and quantities helped us stabilize the residual plot. At this experiment's attribute selection step, the model's initial failure due to the outlier- "satisfying" - rejection and the selecting data failure pivoted this experiment on the right track.

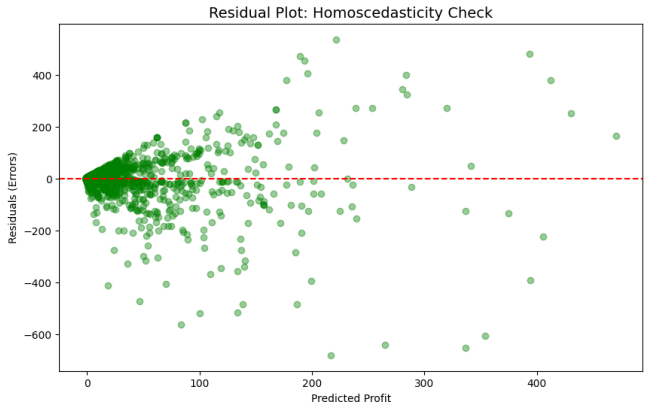

Fig. 6. Residual Plot: Homoscedasticity Check Chart

VI. Conclusion and Reflection

A. Key Findings

We discovered that factors determining profitability in retail are mainly dependent on the volume of sales generated. However, there is a significant vulnerability towards the rate of discounts that are offered. With our Multiple Linear Regression model, we were able to obtain a more solid and precise framework for estimation, attaining an R^2 of 0.359.

B. Experience with Real-World Data

Working on the Superstore dataset using Linear Regression illustrated that in data science the majority of the work can be targeted toward the preparatory work and only a small fraction is focused on modelling. The most significant learning experience for me is the recognition that a result that in isolation may look bad (e.g. an initial negative R^2) may be a sign of the data cleaning phase of the work requiring more time.

C. Challenges and Future Strategies

The largest obstacle encountered was Heteroscedasticity, where the errors "fan out" at greater sales values. Although MLR made some improvements, heteroscedasticity was still present. For future assignments, I would consider Logarithmic Transformations or Polynomial Regression to better address the non-linear relationship at large sales volume and profit margin.